APM1110_FA8_Paglinawan_ZeusAlexis

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Problem 1 – Urn Problem

Given:

- Red balls = 10
- Black balls = 8
- Green balls = 7
- Total balls = 25
- Number of selected balls = 4
- Each color has an equal probability of being chosen.

Let:

X = Number of selected balls that match the chosen color.

1(a) Find P(X = 0)

Solution:

Since each color is equally likely to be chosen,

$$P(X=0) = \frac{1}{3}[P(X=0|R) + P(X=0|B) + P(X=0|G)]$$

If red is chosen:

$$P(X=0|R) = \frac{\binom{15}{4}}{\binom{25}{4}} = \frac{1365}{12650}$$

If black is chosen:

$$P(X=0|B) = \frac{\binom{17}{4}}{\binom{25}{4}} = \frac{2380}{12650}$$

If green is chosen:

$$P(X=0|G) = \frac{\binom{18}{4}}{\binom{25}{4}} = \frac{3060}{12650}$$

Therefore,

$$P(X=0) = \frac{1}{3}\left[\frac{1365+2380+3060}{12650}\right]$$

$$= \frac{6805}{37950}$$

Final Answer:

$$\boxed{P(X=0) = 0.1793}$$

1(b) Find P(X_i = 1), i = 1, 2, 3, 4

Given:

- Red balls = 10
- Black balls = 8
- Green balls = 7
- Total balls = 25

Let:

X_i = 1 if the ith selected ball matches the chosen color, and 0 otherwise.

Solution:

Using the law of total probability,

$$P(X_i = 1) = \frac{1}{3}[P(X_i = 1|R) + P(X_i = 1|B) + P(X_i = 1|G)]$$

For red:

$$P(X_i = 1|R) = \frac{10}{25}$$

For black:

$$P(X_i = 1|B) = \frac{8}{25}$$

For green:

$$P(X_i = 1|G) = \frac{7}{25}$$

Substituting,

$$P(X_i = 1) = \frac{1}{3}\left(\frac{10}{25} + \frac{8}{25} + \frac{7}{25}\right)$$

$$= \frac{1}{3}\left(\frac{25}{25}\right)$$

Final Answer:

$$\boxed{P(X_i = 1) = 0.3333, \quad i = 1, 2, 3, 4}$$

1(c) Find E[X]

Given:

X = X_1 + X_2 + X_3 + X_4

Required:

Find the expected value of X.

Solution:

Express X in terms of the indicator variables:

$$X = X_1 + X_2 + X_3 + X_4$$

Using the linearity of expectation,

$$E[X] = E[X_1 + X_2 + X_3 + X_4]$$

$$= E[X_1] + E[X_2] + E[X_3] + E[X_4]$$

Since each variable is an indicator variable,

$$E[X_i] = P(X_i = 1)$$

Therefore,

$$E[X] = 4\left(\frac{1}{3}\right)$$

Final Answer:

$$\boxed{E[X] = 1.3333}$$

Problem 2 – Yule-Simon Distribution

Given:

The probability mass function is:

$$P(X = n) = \frac{4}{n(n+1)(n+2)}, \quad n \geq 1$$

2(a) Show that the PMF sums to 1

Required:

Show that:

$$\sum_{n=1}^{\infty} P(X = n) = 1$$

Solution:

For a valid probability mass function,

$$\sum_{n=1}^{\infty} P(X = n) = 1$$

Substituting the given PMF,

$$\sum_{n=1}^{\infty} \frac{4}{n(n+1)(n+2)}$$

Using the given identity,

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Therefore,

$$= 2\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Expanding,

$$= 2\left[\left(\frac{1}{2} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{12}\right) + \left(\frac{1}{12} - \frac{1}{20}\right) + \cdots\right]$$

The series is telescoping, so the intermediate terms cancel.

$$= 2\left(\frac{1}{2} - 0\right)$$

Final Answer:

$$\boxed{\sum_{n=1}^{\infty} P(X = n) = 1}$$

Therefore, the given function is a valid PMF.

2(b) Show that E[X] = 2

Given:

$$P(X = n) = \frac{4}{n(n+1)(n+2)}$$

Required:

Show that:

$$E[X] = 2$$

Solution:

The expected value of a discrete random variable is

$$E[X] = \sum_{n=1}^{\infty} nP(X = n)$$

Substituting the given PMF,

$$E[X] = \sum_{n=1}^{\infty} n \left[\frac{4}{n(n+1)(n+2)} \right]$$

Canceling n,

$$= 4 \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$$

Using partial fractions,

$$\frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$$

Therefore,

$$E[X] = 4 \sum_{n=1}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+2} \right)$$

Expanding,

$$= 4 \left[\left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \cdots \right]$$

The intermediate terms cancel.

$$= 4 \left(\frac{1}{2} - 0 \right)$$

Final Answer:

$$\boxed{E[X] = 2}$$

2(c) Show that E[X^2] = infinity

Given:

$$P(X = n) = \frac{4}{n(n+1)(n+2)}$$

Required:

Show that:

$$E[X^2] = \infty$$

Solution:

The second moment is given by

$$E[X^2] = \sum_{n=1}^{\infty} n^2 P(X = n)$$

Substituting the given PMF,

$$E[X^2] = \sum_{n=1}^{\infty} n^2 \left[\frac{4}{n(n+1)(n+2)}\right]$$

Simplifying,

$$= 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)}$$

Using partial fractions,

$$\frac{n}{(n+1)(n+2)} = -\frac{1}{n+1} + \frac{2}{n+2}$$

Therefore,

$$E[X^2] = 4\sum_{n=1}^{\infty} \left(-\frac{1}{n+1} + \frac{2}{n+2}\right)$$

Rearranging,

$$= 4\sum_{n=1}^{\infty} \left(\frac{1}{n+2} - \frac{1}{(n+1)(n+2)}\right)$$

The harmonic series diverges:

$$\sum_{n=1}^{\infty} \frac{1}{n+2} = \infty$$

Meanwhile, the second series converges.

Final Answer:

$$\boxed{E[X^2] = \infty}$$

Problem 3 – Continuous Probability Distribution

Given:

The probability density function is:

$$f(x) = \begin{cases} a + bx^2, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

The expected value is:

$$E[X] = \frac{3}{5}$$

Required:

Find the values of a and b.

Solution:

Since the total area under a PDF must equal 1,

$$\int_0^1 (a + bx^2)\, dx = 1$$

Integrating,

$$\left[ax + \frac{bx^3}{3}\right]_0^1 = 1$$

Substituting the limits,

$$a + \frac{b}{3} = 1$$

Solving for a,

$$a = 1 - \frac{b}{3} \tag{1}$$

Using the expected value formula,

$$E[X] = \int_0^1 x f(x)\, dx$$

Substituting,

$$\frac{3}{5} = \int_0^1 x(a + bx^2)\, dx$$

Distributing,

$$\frac{3}{5} = \int_0^1 (ax + bx^3)\, dx$$

Integrating,

$$\frac{3}{5} = \left[\frac{ax^2}{2} + \frac{bx^4}{4}\right]_0^1$$

Therefore,

$$\frac{3}{5} = \frac{a}{2} + \frac{b}{4} \tag{2}$$

Substituting equation (1) into equation (2),

$$\frac{3}{5} = \frac{1}{2}\left(1 - \frac{b}{3}\right) + \frac{b}{4}$$

Distributing,

$$\frac{3}{5} = \frac{1}{2} - \frac{b}{6} + \frac{b}{4}$$

Combining like terms,

$$\frac{3}{5} = \frac{1}{2} + \frac{b}{12}$$

Moving the constant,

$$\frac{3}{5} - \frac{1}{2} = \frac{b}{12}$$

$$\frac{1}{10} = \frac{b}{12}$$

Solving for b,

$$b = \frac{12}{10} = 1.2$$

Substituting b into equation (1),

$$a = 1 - \frac{1.2}{3}$$

$$a = 1 - 0.4$$

$$a = 0.6$$

Final Answers:

$$\boxed{a = 0.6000}$$

$$\boxed{b = 1.2000}$$

Problem 4 – Standardization of Random Variables

Given:

A random variable X has:

$$E[X] = \mu$$

$$Var(X) = \sigma^2$$

The standardized random variable is:

$$Y = \frac{X - \mu}{\sigma}$$

Required:

Find:

1. The expected value of Y.
2. The variance of Y.

Solution:

Expected Value:

Using the expected value formula,

$$E[Y] = E\left[\frac{X - \mu}{\sigma}\right]$$

Since sigma is a constant,

$$E[Y] = \frac{1}{\sigma} E[X - \mu]$$

Applying linearity of expectation,

$$E[Y] = \frac{1}{\sigma}(E[X] - \mu)$$

Since E[X] = mu,

$$E[Y] = \frac{1}{\sigma}(\mu - \mu)$$

$$\boxed{E[Y] = 0}$$

Variance:

Using the variance property,

$$Var(aX + b) = a^2 Var(X)$$

Given,

$$Y = \frac{X}{\sigma} - \frac{\mu}{\sigma}$$

Therefore,

$$Var(Y) = \frac{1}{\sigma^2} Var(X)$$

Since Var(X) = sigma squared,

$$Var(Y) = \frac{1}{\sigma^2}(\sigma^2)$$

$$\boxed{Var(Y) = 1}$$

Problem 5 – Sum of Independent Poisson Random Variables

Given:

X and Y are independent Poisson random variables.

$$X \sim Poisson(\lambda_1 = 2)$$

$$Y \sim Poisson(\lambda_2 = 3)$$

Let:

$$W = X + Y$$

5(a) Derive the MGF M_W(t) of W

Required:

Derive the moment-generating function of W.

Solution:

The moment-generating function of a Poisson random variable is

$$M_X(t) = e^{\lambda(e^t - 1)}$$

For X, where lambda = 2,

$$M_X(t) = e^{2(e^t-1)}$$

For Y, where lambda = 3,

$$M_Y(t) = e^{3(e^t-1)}$$

Since X and Y are independent,

$$M_W(t) = M_X(t)M_Y(t)$$

Substituting,

$$M_W(t) = e^{2(e^t-1)}e^{3(e^t-1)}$$

Combining the exponents,

$$M_W(t) = e^{2(e^t-1)+3(e^t-1)}$$

Simplifying,

$$\boxed{M_W(t) = e^{5(e^t-1)}}$$

5(b) Identify the probability distribution and parameter(s) of W

Given:

$$M_W(t) = e^{5(e^t-1)}$$

Required:

Identify the probability distribution and its parameter.

Solution:

The MGF of a Poisson random variable is

$$M(t) = e^{\lambda(e^t-1)}$$

From part (a),

$$M_W(t) = e^{5(e^t-1)}$$

Comparing the two expressions,

$$\lambda = 5$$

Therefore, using the uniqueness property of MGFs,

$$\boxed{W \sim Poisson(5)}$$

The parameter is

$$\boxed{\lambda_W = \lambda_1 + \lambda_2 = 2 + 3 = 5}$$

5(c) Determine E[W] and E[Wˆ2]

Given:

$$M_W(t) = e^{5(e^t - 1)}$$

Required:

Find:

$$E[W] = M'_W(0)$$

$$E[W^2] = M''_W(0)$$

Solution:

Given the MGF,

$$M_W(t) = e^{5(e^t - 1)}$$

The first derivative gives the expected value:

$$E[W] = M'_W(0)$$

The second derivative gives the second moment:

$$E[W^2] = M''_W(0)$$

First Derivative:

Using the chain rule,

$$M'_W(t) = e^{5(e^t - 1)}(5e^t)$$

Rearranging,

$$M'_W(t) = 5e^t e^{5(e^t - 1)}$$

Evaluating at t = 0,

$$M'_W(0) = 5e^0 e^{5(e^0 - 1)}$$

Since eˆ0 = 1,

$$M'_W(0) = 5(1)e^0$$

Therefore,

$$\boxed{E[W] = 5.0000}$$

Second Derivative:

Using the product rule,

$$M''_W(t) = 5e^t e^{5(e^t-1)} + 5e^t(5e^t e^{5(e^t-1)})$$

Simplifying,

$$M''_W(t) = 5e^t e^{5(e^t-1)} + 25e^{2t} e^{5(e^t-1)}$$

Factoring out the common term,

$$M''_W(t) = (5e^t + 25e^{2t})e^{5(e^t-1)}$$

Evaluating at t = 0,

$$M''_W(0) = (5e^0 + 25e^0)e^{5(e^0-1)}$$

Simplifying,

$$M''_W(0) = (5 + 25)(1)$$

$$M''_W(0) = 30$$

Therefore,

$$\boxed{E[W^2] = 30.0000}$$